Supplementary Tables for CODA-SC

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Table S1: Per-workload results for representative methods. Calibrated exits answer most requests locally on every workload; CODA-SC's controller removes the residual deadline misses that conformal-exit incurs on offloaded requests.

Dataset	Method	Acc. (%)	Mean ms	Miss (%)	Upload KB	Exit (%)
MNIST	Cloud only	99.66	108.3	54.8	1.00	0.0
MNIST	Confidence exit	98.63	19.6	0.0	0.00	100.0
MNIST	AODPart-style	98.63	19.6	0.0	0.00	100.0
MNIST	Conformal exit	99.54	24.8	2.7	0.05	95.0
MNIST	CODA-SC-Cov	99.54	22.2	0.1	0.01	95.0
MNIST	CODA-SC-Risk	99.50	21.6	0.0	0.00	99.2
Fashion-MNIST	Cloud only	94.16	108.1	50.8	1.00	0.0
Fashion-MNIST	Confidence exit	92.51	22.6	0.2	0.00	99.7
Fashion-MNIST	AODPart-style	92.51	22.4	0.0	0.00	99.7
Fashion-MNIST	Conformal exit	90.81	20.8	0.4	0.01	99.1
Fashion-MNIST	CODA-SC-Cov	90.81	20.3	0.0	0.00	99.1
Fashion-MNIST	CODA-SC-Risk	93.94	28.2	0.1	0.01	94.1
CIFAR-10	Cloud only	88.12	110.2	52.4	3.00	0.0
CIFAR-10	Confidence exit	87.20	42.9	6.2	0.36	88.1
CIFAR-10	AODPart-style	87.20	36.5	0.2	0.09	88.1
CIFAR-10	Conformal exit	86.04	34.9	1.2	0.07	97.7
CIFAR-10	CODA-SC-Cov	86.04	33.7	0.0	0.02	97.7
CIFAR-10	CODA-SC-Risk	87.98	43.1	0.2	0.11	79.3
SVHN	Cloud only	96.05	111.0	48.6	3.00	0.0
SVHN	Confidence exit	93.84	30.4	0.3	0.02	99.4
SVHN	AODPart-style	93.84	30.1	0.0	0.01	99.4
SVHN	Conformal exit	93.95	32.8	1.8	0.11	96.3
SVHN	CODA-SC-Cov	93.95	30.8	0.0	0.03	96.3
SVHN	CODA-SC-Risk	95.54	35.3	0.1	0.04	95.3

Table S2: Residual action rates by network profile for AODPart-style and CODA-SC policies. Rates are averaged over all requests; most requests have already terminated at a local exit, so the nonzero entries describe the small uncertain residual stream served by the controller.

Method	Profile	Local (%)	Raw (%)	Split1 (%)	Split2 (%)	Split3 (%)	Offload (%)
AODPart-style	Wifi	0.00	3.21	0.00	0.00	0.00	3.21
AODPart-style	Lte	3.11	0.10	0.00	0.00	0.00	0.10
AODPart-style	Congested	3.21	0.00	0.00	0.00	0.00	0.00
AODPart-style	Mixed	3.21	0.01	0.00	0.00	0.00	0.01
CODA-SC-Cov	Wifi	0.00	2.97	0.00	0.00	0.00	2.97
CODA-SC-Cov	Lte	2.96	0.01	0.00	0.00	0.00	0.01
CODA-SC-Cov	Congested	2.97	0.00	0.00	0.00	0.00	0.00
CODA-SC-Cov	Mixed	2.97	0.00	0.00	0.00	0.00	0.00
CODA-SC-Risk	Wifi	1.74	6.28	0.00	0.00	0.00	6.28
CODA-SC-Risk	Lte	8.01	0.01	0.00	0.00	0.00	0.01
CODA-SC-Risk	Congested	8.02	0.00	0.00	0.00	0.00	0.00
CODA-SC-Risk	Mixed	8.02	0.00	0.00	0.00	0.00	0.00

Table S3: Sensitivity of CODA-SC to the conformal level α and the edge throughput Φ_e (GFLOP/s).

Sweep	Value	Exit (%)	Miss (%)	Mean ms	Sel. err. (%)
α	0.05	95.0	0.1	30.1	4.49
α	0.1	97.0	0.1	26.8	6.69
α	0.15	95.6	0.1	25.0	8.27
α	0.2	93.3	0.1	24.7	8.88
Edge GFLOP/s	2.5	97.0	26.3	52.9	6.69
Edge GFLOP/s	5	97.0	0.1	26.8	6.69
Edge GFLOP/s	10	97.0	0.0	13.5	6.69
Edge GFLOP/s	20	97.0	0.0	6.7	6.69

Table S4: Payload-scaling split-selection diagnostic. Raw-input upload is scaled by 128×, intermediate activation payloads by 0.125×, and exits are disabled in the controller rows to isolate the split/fallback action space. This is a payload-regime diagnostic rather than learned-codec or hardware evidence.

Method	Mean ms	P95 ms	Miss (%)	Upload KB	Local (%)	Raw (%)	Split1 (%)	Split2 (%)	Split3 (%)
Cloud only	402.6	934.8	91.02	256.00	0.0	100.0	0.0	0.0	0.0
Deadline greedy	124.9	267.1	65.09	2.00	0.0	0.0	100.0	0.0	0.0
Oracle split	124.9	267.1	65.08	2.90	0.0	0.7	99.3	0.0	0.0
CODA split controller (no exits)	124.9	267.1	65.09	2.00	0.0	0.0	100.0	0.0	0.0
CODA controller (no exits)	61.0	61.0	0.03	0.00	99.8	0.0	0.2	0.0	0.0